STATISTICAL ARBITRAGE

Machine Learning for Sports Betting

Accuracy, calibration and trading performance in the NBA and English Premier League

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Final project report

Research question. Does selecting a prediction model for calibration, rather than accuracy, improve the performance of an explicit sports-betting rule? We implement the same research framework in a binary NBA market and a three-outcome Premier League market, while preserving their different data, selection dates and capital-allocation conventions.

Main findings. In the locked EPL experiment neither branch selects an admissible validation trading rule, so both primary test policies hold cash. Its calibration branch's secondary flat-stake yield is 0.41%, with an interval crossing zero. New EPL player models do not improve matched team controls. The reproducible NBA logistic account returns 0.98% with flat stakes and -35.00% with eighth-Kelly; separately restarted annual logistic accounts have negative average returns for both objectives. The source-reported SVM returns 25.03% flat but -55.11% with Kelly; its refit does not reproduce. Bankrupt-policy reversals are reported separately as post-hoc sensitivity tests, not confirmed profitable policies.

Evidence standard. Original saved outputs are distinguished from independent calculations. The NBA logistic replay reconciles to the supplied bankrolls; the SVM refit does not, and its original numbers remain explicitly source-reported. Historical profits are not presented as executable or prospectively validated investment returns.

Contributions: Aman Dhillon — NBA; Panagiotis Housos — Premier League.
Order: common methodology; NBA results; Premier League results.
Companions: editable presentation, complete LaTeX source, numerical tables, strategy definitions and source/output hash manifest.

1 Common methodology

1.1 Research question and experimental scope

The September 30 proposal asks whether selecting a sports-outcome model for calibration, rather than classification accuracy, improves betting returns. Aman Dhillon implements the NBA component and Panagiotis Housos the English Premier League (EPL) component. Both studies connect pre-game statistical features to outcome probabilities and then to explicit betting rules. A good classifier, a well-calibrated probability forecast and a profitable betting strategy are different objects: a profitable bet also requires a sufficiently favourable price and a viable stake.

The motivating reference is Walsh and Joshi's study of NBA model selection [1]. Its 2025 corrigendum supersedes the original numerical claims: corrected mean capital returns are -9.77% for calibration-based selection and -26.78% for accuracy-based selection, and Kelly betting can lose despite calibration [2]. Consequently, improved profitability is tested rather than assumed. The NBA experiment is a replication and rolling extension; the EPL experiment adapts the pipeline to three outcomes and subsequently adds exploratory team, xG and player-feature comparisons. Multi-book line shopping and mathematical arbitrage screening are implemented in the EPL component, not the supplied NBA notebook's single-book replay.

1.2 Pre-game information and chronological model selection

For fixture i, let Y_i denote the realised result and X_i the feature vector constructed from earlier completed games. NBA results are binary, $Y_i \in \{H, A\}$; EPL results include draws, $Y_i \in \{H, D, A\}$. Team features describe earlier opponent-relative performance and prior-season strength. The NBA code applies shifted expanding averages within each team-season. EPL features include season-to-date and recent form, with source-coverage controls and, in the extensions, lagged xG and FPL player aggregates. Target-game scores, appearances and minutes cannot be predictors of that same game.

The initial experiments use separate accuracy- and calibration-driven branches, each considering logistic regression, random forests, support vector machines and multilayer perceptrons. Distribution screening, Spearman redundancy filtering, sequential forward selection and TPE hyperparameter search precede the later betting evaluation. These are shared methodological ideas, not identical settings. All normalisation statistics are estimated in the appropriate earlier fitting block. Table 1 records differences that matter for interpreting the results.

Selecting a model for small calibration error is not the same as fitting a post-processing calibrator. In the first EPL study, a scalar temperature is fitted on a disjoint season for every candidate in both branches, $p_{ik}(T) = \exp(z_{ik}/T)/\sum_j \exp(z_{ij}/T)$, where z_i are the saved model scores [3]. The NBA notebook instead selects using its calibration criterion. Neither implementation implies that probability calibration guarantees value against bookmaker prices.

1.3 Forecast scores and market benchmarks

Let p_{ik} be the predicted probability of outcome k and $y_{ik} = \mathbb{1}\{Y_i = k\}$. Classification accuracy uses the highest-probability outcome. Probability quality is assessed using log loss and Brier loss:

$$\mathrm{LL} = -\frac{1}{n}\sum_i \log p_{i,Y_i}, \qquad \mathrm{Brier}_{\mathrm{EPL}} = \frac{1}{n}\sum_i \sum_{k\in\{H,D,A\}} (p_{ik} - y_{ik})^2. \tag{1}$$

The NBA binary Brier convention is $n^{-1}\sum_i (p_{iH} - y_{iH})^2$. Its values therefore should not be ranked directly against the unscaled three-class EPL Brier score. Log-loss levels also depend on the outcome space and base rates; model–market differences are interpreted within each league.

Table 1: Shared framework and league-specific implementation.

Aspect	NBA	Premier League
Outcome and feature eligibility	Home/away; both teams must have at least ten earlier season games.	Home/draw/away; missing histories retained with coverage and fitting-only missing-value handling.
First selection chronology	Train 2014/15–2015/16; tune 2016/17; choose model family on 2017/18; bet 2018/19.	Train 2014/15–2020/21; tune 2021/22; calibrate 2022/23; select family 2023/24; choose threshold 2024/25; evaluate 2025/26.
Calibration treatment	Selection by a 20-bin binary ECE with an occupancy penalty; no separate held-out temperature fit.	Selection by mean three-class ECE with ten bins; both branches receive held-out temperature scaling.
Extensions	Nine rolling windows reselect features but fit logistic regression in both branches.	Four rolling evaluation seasons compare fixed feature groups; the later player-model study adds logistic and boosted trees.
Prices and outcome choice	One saved home/away moneyline pair; test home value first, otherwise away value.	Saved individual-book quotes and a Bet365 reference; select the allowed outcome with maximum estimated return.
Capital allocation	Sequential per-game settlement; fixed 100 or one-eighth Kelly from 10,000 initial capital.	Same-date opening-cash allocation; fixed, cash-fraction and capped fractional-Kelly configurations; 100,000 initial capital in the expanded account comparison.

For probability bin b of class k, calibration error compares observed frequency $\bar{y}_{kb}$ with mean forecast $\bar{p}_{kb}$:

$$\mathrm{ECE}_k = \sum_b \frac{n_{kb}}{n} |\bar{y}_{kb} - \bar{p}_{kb}|. \tag{2}$$

The EPL implementation averages this quantity over all three classes and reports ten-bin ECE, with five- and fifteen-bin sensitivity summaries. The NBA implementation uses the home-win probability and twenty equal-width bins; it replaces the score by 1 if fewer than 80% of bins are occupied. This is an optimisation penalty, not an observation of literal 100% miscalibration. The two ECE scales are not interchangeable. Constant forecasts may have small unpenalised ECE while offering little match-specific information.

For decimal odds O_{ik}, the implied probability is $1/O_{ik}$. A reference book's overround and proportional margin-normalised benchmark are

$$A_i = \sum_k O_{ik}^{-1}, \qquad q_{ik} = \frac{O_{ik}^{-1}}{A_i}. \tag{3}$$

The benchmark q_i is a market estimate, not known truth. NBA American moneylines m are converted to decimal odds as $1 + m/100$ for $m > 0$, and $1 + 100/|m|$ for $m < 0$. EPL line shopping uses the largest eligible archived individual-book quote for each outcome; a separate reference-price replay recomputes selections and stake sizes. An archived combination with $\sum_k 1/O_{ik}^* < 1$ is a mathematical arbitrage candidate only. Without simultaneous accepted fills, aligned settlement and known costs, it is not an executed risk-free profit.

1.4 Bet selection, stake sizing and accounting

For selected outcome probability p_i and decimal odds O_i, estimated return per unit staked is $e_i = p_i O_i - 1$. The NBA rule takes a home bet when $p_{iH} > 1/O_{iH}$, otherwise an away bet when $1 - p_{iH} > 1/O_{iA}$; it does not

maximise across both qualifying sides. The principal EPL value rules instead choose the largest admissible e_i and require a declared threshold. Thresholds and filters are reported with their respective strategies; the first EPL validation-selected rules can remain entirely in cash.

With stake s_i, the realised net payoff and the model-implied positive and negative cash components are

$$\pi_i = s_i\{O_i \mathbb{1}(\text{win}) - 1\}, \tag{4}$$

$$G_i = p_i s_i (O_i - 1), \qquad L_i = (1 - p_i)s_i, \qquad G_i - L_i = s_i e_i. \tag{5}$$

Expected gains and losses use model probabilities at the recorded stakes; they are distinct from average realised winning gains and losing stakes. They are reported only where the supplied prediction-and-trade evidence supports their calculation.

Full Kelly requests fraction $f_i^* = (p_i O_i - 1)/(O_i - 1)$ of available capital. Fractional Kelly multiplies this by λ, with any declared per-bet cap c: $s_i = B \min\{c, \lambda \max(0, f_i^*)\}$. The NBA notebook uses $\lambda = 1/8$ and caps only at available cash. EPL configurations specify their Kelly fraction, per-bet cap and any total date-level budget. A fixed stake of 1,000 in a 100,000 account equals 1% of *initial* capital, not 1% of a changing bankroll. EPL same-date requests are reduced together to fit opening cash before any same-date winnings settle. The NBA source simulator processes games sequentially, so its funding convention is retained and disclosed rather than silently replaced by the EPL convention.

For total profit $P = \sum_i \pi_i$ and turnover $V = \sum_i s_i$, the final report uses two explicitly different denominators:

$$\text{stake ROI (yield)} = P/V, \qquad \text{capital return} = P/B_0. \tag{6}$$

The supplied NBA material calls the second quantity "ROI"; it is relabelled as capital return for consistency. Hit ratio is wins divided by settled bets; trade frequency is bets divided by priced match opportunities. Drawdown is the largest proportional fall from a prior account peak. Undefined no-trade ratios remain unavailable. Where a display rebases an account to 100,000, all cash amounts and stakes are multiplied by the same constant; reported percentage returns do not change. Separately reset NBA seasons are not presented as one funded continuous investment account.

1.5 Interpretation and evidence limits

The completed historical experiments are comparisons of recorded designs, not new live trading records. Repeatedly examined EPL extensions are explicitly exploratory. NBA annual variation and EPL whole-date conditional bootstrap intervals do not remove feature, model or strategy selection uncertainty. Bookmaker probabilities are competitive benchmarks, and favourable model expected value is not independent evidence of a realised edge. Neither a positive season nor a historical rank establishes a reliable betting policy. The NBA results below preserve the supplied notebook and report evidence; the EPL section retains the frozen prediction, trade and cash-flow artifacts.

2 NBA results

2.1 Data, features and selection roles

The NBA component uses the supplied team-game CSV and season tables in `OddsData.sqlite`. The CSV contains 33,316 team records and 16,658 distinct raw games across 2010/11–2023/24. After pairing teams and excluding each club's first ten season appearances, 14,471 games remain; both teams have at least ten earlier observations in each retained fixture. The core betting season, 2018/19, contains 1,074 eligible games with a complete matched moneyline pair. The saved 58.5% home-win rate refers to the training subset, not to all 14,471 games.

The initial feature set consists of 25 opponent-relative box-score statistics and a previous-season win-rate difference. Shifted expanding means exclude the target game. The training–validation KS screen removes offensive rebounds; 25 candidates remain. The Spearman screen retains nine variables. Forward selection uses offensive rating and three-pointers made for the calibration branch, and adds prior-season win percentage and turnovers for the accuracy branch. These are different predictor subsets, not a single model scored two ways.

Table 2: NBA chronology and sample sizes in the supplied core experiment.

Role	Seasons	Games
Initial training	2014-15, 2015-16	2,145
Feature / hyperparameter validation	2016-17	1,074
Model-family selection	2017-18	1,075
Betting evaluation	2018-19	1,074

There are 30 Optuna trials per family and branch. The source averages predictions over ten random seeds for model-family assessment and the final betting prediction. The season called *test* in the notebook is used to select the family: it is therefore a model-selection season in this report. Only the subsequent 2018/19 betting season follows all these choices.

Table 3: NBA model-family selection, 2017/18. The two columns describe different branch-specific fitted variants; they are not joint scores of one model.

Family	B: ECE (2 inputs)	C: accuracy (4 inputs)
Logistic regression	4.31%	65.30%
Random forest	5.68%	65.40%
Neural network	4.65%	65.49%
SVM	4.40%	66.33%

Calibration selection chooses logistic regression; accuracy selection chooses the SVM. The calibration branch's selection ECE is 4.31%, while the accuracy branch's selection accuracy is 66.33%. They must not be described as the accuracy and ECE of a single predictor. *Evidence:* `MLSports/Data.ipynb: cells1--16(zero-based)`

2.2 The 2018/19 betting simulation

Each native account starts with 10,000 monetary units. Flat staking requests 100 per qualifying game, equal to 1% of *initial* capital. One-eighth Kelly varies the request with the predicted edge and available cash; unlike the EPL rules, no additional 2% per-bet ceiling is imposed. The simulator checks home value before away

value and settles games sequentially. Its output labelled "ROI" is profit divided by starting capital. Table 4 uses the unambiguous term *capital return*.

Table 4: Original saved NBA betting results, 2018/19; initial capital 10,000.

Branch / model	Stake	Final capital	Capital return	Bet freq.	Hit ratio
Calibration / LR	Flat 100	10,097.74	0.98%	85.3%	38.0%
Calibration / LR	1/8 Kelly	6,500.40	-35.00%	85.3%	38.0%
Accuracy / SVM	Flat 100	12,502.98	25.03%	88.6%	34.9%
Accuracy / SVM	1/8 Kelly	4,488.59	-55.11%	88.6%	34.9%

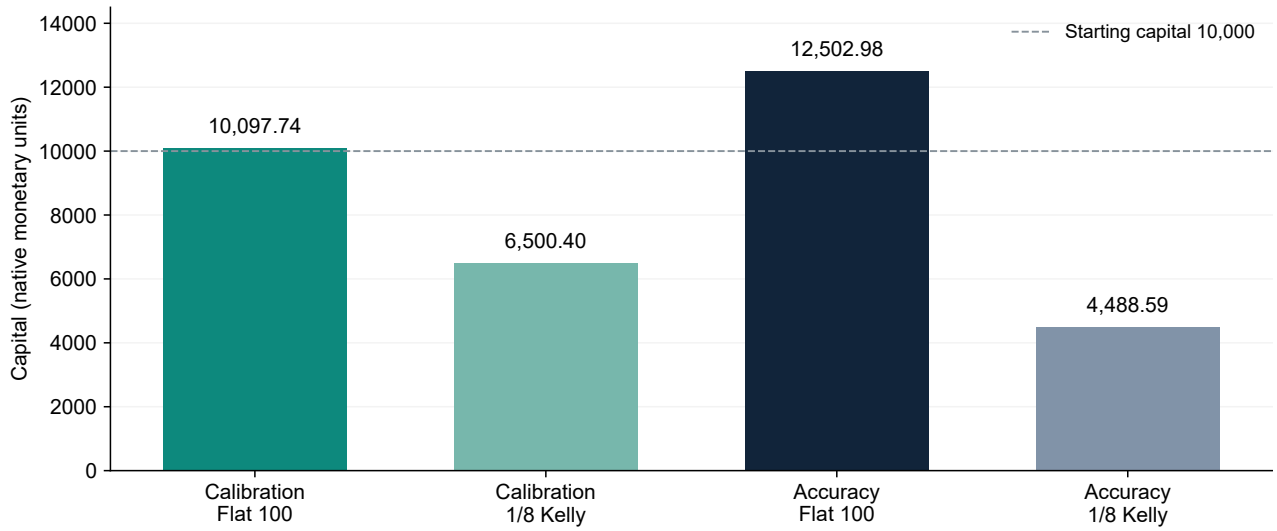


Figure 1: Ending capital of the four original saved NBA 2018/19 strategies. These are endpoint comparisons, not a reconstructed SVM wealth path. LR = logistic regression. The SVM numbers are source-reported; their supplied-parameter refit does not reconcile.

The accuracy-selected SVM's positive flat-stake result and negative Kelly result come from different exposure sizes, not different prediction models. For the calibration-selected logistic model, flat stakes finish only slightly above the starting account, while Kelly finishes substantially below it. Thus the source evidence does not support treating calibration selection as a guarantee of profitable stake sizing.

The saved betting-season prediction accuracy is 65.27% for logistic regression and 64.80% for the SVM, distinct from their earlier selection scores. The notebook's SVM calibration score is replaced by a bin-coverage penalty; it is not an observation of literal 100% calibration error.

2.3 What the independent replay verifies

A fixed-parameter audit uses the supplied feature subsets and stored hyperparameters with ten-seed averaging. It performs no new feature search, hyperparameter search or strategy optimisation. Both logistic account endings match the notebook to the printed cent, and its saved accuracy and ordinary 20-bin ECE also reconcile. In contrast, the SVM refits reach the supplied iteration limit and produce different predictions and bankrolls. Original library versions and serialized fitted weights were not supplied. We retain the original SVM results as *reported* and do not silently replace them with the mismatching refit.

Table 5: Independently computed metrics on the same 1,074 NBA betting fixtures. The logistic reconstruction reconciles; the bookmaker benchmark is calculated directly from the provided odds.

Predictor	Accuracy	ECE 20	Log loss	Binary Brier
Matched logistic	65.27%	4.27%	0.6172	0.2141
Normalised odds	66.95%	4.73%	0.5954	0.2045

The market probabilities have lower log loss and binary Brier loss and higher accuracy than the reconciled logistic predictor. The logistic model, however, has lower ordinary ECE in this season. We therefore do not claim that the market dominates every metric in every season. In particular, a small global calibration statistic does not establish a profitable advantage on the bets selected against bookmaker prices.

For the same reconciled flat LR account, mean selected-outcome probability is 45.7821% versus observed hit rate 37.9913% (a 7.7909 percentage-point gap), despite ordinary global ECE 4.2721%. Model-implied net payoff is 29.487 units/trade; realised net is 0.107. A small aggregate calibration error need not transfer to the price-selected bet subset.

Table 6: Additional metrics supported by the cent-matched NBA logistic replay; native 10,000-unit account. Expected components use the model's probabilities, not observed win/loss averages.

Stake	Bets	Hit	Freq.	G/trade	L/trade	Stake ROI	Max DD
Flat 100	916	37.99%	85.29%	83.70	54.22	0.11%	29.09%
1/8 Kelly	916	37.99%	85.29%	103.43	54.37	-3.26%	59.45%

Only the reconciled logistic subset supports these extra trade-ledger statistics. Corresponding SVM turnover, expected payoff components and drawdown are not imputed from the original rounded headline outputs. *Evidence:* `final_deliverables/nba_core_replay/summary.json`

2.4 Nine independently restarted seasons

The rolling extension contains nine betting seasons, 2014/15–2022/23, and 9,344 priced games. It reselects features in each window, but uses logistic regression for *both* branches. It does not repeat the core four-family hyperparameter tournament. Each seasonal diagnostic restarts at 10,000 and requests flat stakes of 100.

Table 7: Saved NBA annual capital returns, flat 100 from a fresh 10,000 each season. Values are preserved at the source's two-decimal precision.

Season	Games	B inputs	C inputs	Cal. capital return	Acc. capital return
2014-15	1,072	2	7	-65.30%	-16.78%
2015-16	1,073	5	4	-17.73%	-12.31%
2016-17	1,074	4	4	57.42%	46.75%
2017-18	1,075	4	6	-100.00%	-100.00%
2018-19	1,074	2	4	2.08%	12.25%
2019-20	906	5	4	85.58%	100.86%
2020-21	924	6	6	-31.59%	-33.85%
2021-22	1,073	3	7	-83.67%	-89.28%
2022-23	1,073	3	3	-2.89%	6.93%

The calibration branch has the higher return in only three of nine seasons. Its reported mean return is

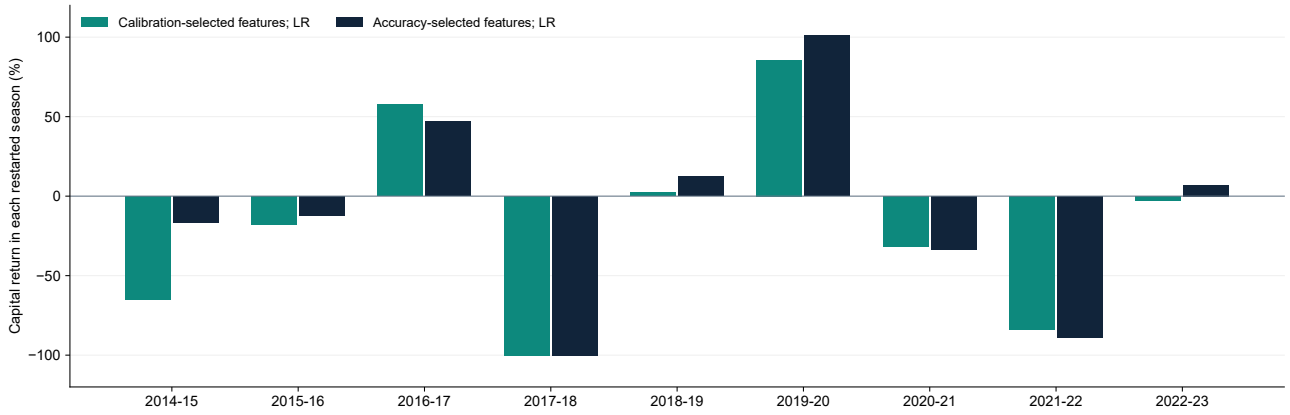


Figure 2: NBA annual capital returns for the rolling logistic-only extension. Every bar represents a separate initial 10,000 account; the bars are not contributions to a continuously funded portfolio.

-17.34% with sample standard deviation 61.7 percentage points, versus -9.49% and 62.5 points for the accuracy branch. Both 2017/18 accounts exhaust their cash. These nine dependent historical windows are descriptive robustness evidence, not nine independent replications establishing a precise long-run expected return.

The source's cumulative display adds cashflows from separately restarted seasons to a previous base. That can create unfunded or negative shadow balances and is not a continuous self-financing bankroll. It is excluded from this report's money-progression evidence. The supplied NBA write-up also describes additional pooled market/fusion analyses whose underlying forecasts are not preserved in the notebook. Those figures are not substituted for the directly reconciled common-fixture comparison above.

2.5 Opposite bets after bankruptcy: NBA diagnostic

Both zero annual accounts occur in the 2017/18 logistic-only extension, not the four-family 2018/19 experiment. We reconstruct all nine annual windows: all 18 flat-stake returns match the notebook's printed precision. This method-and-endpoint reconciliation does not recover saved fitted weights or prove byte-identical original predictions. Only two of the 36 audited annual flat/Kelly accounts are exactly zero; Kelly leaves positive balances.

For each zero account we keep the original qualifying dates and forecast, back the other team at its archived opposing moneyline, and request 100 (1% of initial 10,000). The inverse continues after the original account stops, clamps each stake to its own cash, and retains source sequential settlement. It is not a negative copy of the original profit.

Table 8: NBA full-horizon opposite bets, 2017/18; original ending capital is zero in both rows. Intervals bootstrap whole dates with fitted predictions, signals and stakes fixed.

Branch	Signals	Bets	End capital	Capital return	Hit	Stake yield	95% interval
Calibration	915	915	11,148.07	11.48%	65.68%	1.25%	[-4.82%, 7.33%]
Accuracy	884	884	8,785.30	-12.15%	63.35%	-1.37%	[-7.36%, 4.53%]

The calibration-direction inverse ends at 11,148.07, while the accuracy-direction inverse ends at 8,785.30. Reversing only the recorded pre-bankruptcy bets instead ends at 9,420.33 and 8,155.98: the observation win-

dow matters. Both full-horizon stake-yield intervals contain zero. These cases were selected after observing ruin; archived opposing quotes and sequential funding are not evidence of executable fills or new alpha.

3 Premier League results

3.1 Data and adaptation to three outcomes

The EPL component predicts home wins, draws and away wins. The focal universe is the 20 saved 2026/27 Premier League clubs; historical opponents remain context rather than being relabelled as current clubs. This present-day cohort creates selection/survivor limitations. Its 2025/26 matched evaluation contains 374 fixtures, rather than all 380 league fixtures.

Football-Data supplies results, team performance and individual-book odds; Understat supplies team expected-goals histories; Fantasy Premier League (FPL) supplies player records. The expanded player loader reconciles 253,509 historical player-match records in 2016/17–2025/26. The saved October 20 snapshot adds 4,551 current-season rows from 70 completed fixtures and 668 current player summaries. Historical inputs use strictly earlier dates; target participants, target minutes and present-day rosters do not backfill earlier forecasts. Role-based extensions disclose season-fixed position proxies with unverified publication clocks.

3.2 Proposal-aligned accuracy versus calibration experiment

The first locked experiment considers the same four candidate families as the NBA component, with 12 seeded tuning trials per family per branch. Earlier training and tuning are followed by separate calibration, model-selection and trading-validation seasons (Table 1). Both branches receive temperature scaling; the experiment compares selection objectives rather than calibrated versus deliberately uncalibrated predictions.

Accuracy selection chooses an SVM using season-to-date corners difference and recent shots-on-target difference. Calibration selection chooses an MLP using earlier corners conceded. The original xG/player candidates do not survive this small-budget selection path, so this experiment alone cannot establish their incremental value.

Table 9: Locked EPL 2025/26 probability scores on 374 identical fixtures. ECE is mean classwise error with ten bins.

Predictor	Accuracy	ECE 10	Log loss	Brier
Accuracy-selected SVM	48.13%	6.20%	1.0763	0.6446
Calibration-selected MLP	44.39%	4.16%	1.0564	0.6356
Normalised Bet365	48.66%	4.36%	1.0191	0.6120
Training frequencies	42.51%	2.90%	1.0851	0.6566

The calibration-selected MLP has lower ECE, log loss and Brier loss, but lower classification accuracy, than the selected SVM. Normalised bookmaker probabilities have stronger proper losses than either learned model. The frequency baseline's low ECE despite poor log loss also illustrates the limitations of using calibration error alone. Conditional paired intervals for the MLP-minus-SVM ECE and log-loss differences both contain zero; they omit model-selection uncertainty and do not establish a causal calibration effect.

Neither primary policy achieves positive validation profit with at least 30 validation bets. Both therefore select cash: zero test bets, zero turnover and undefined stake ROI. This is the primary locked-policy finding. The secondary 5% edge rule is evaluated separately, with an original 1,000-unit account, flat 10 or capped quarter Kelly.

The MLP's best-price flat-stake ROI of 0.41% is not robust evidence of an edge: its interval crosses zero, its reference-price flat result is negative, and its quarter-Kelly result is negative. Both primary rules remain

Table 10: Secondary EPL 5% edge replay on the test season; native initial capital 1,000. Stake ROI is profit divided by turnover.

Branch / odds	Stake	Bets	Profit	Stake ROI	95% ROI interval
SVM / Best	Flat 10	371	-83.30	-2.25%	[-18.83%, 15.45%]
SVM / Best	1/4 Kelly	371	-377.05	-7.01%	[-23.18%, 11.92%]
SVM / Bet365	Flat 10	365	-186.70	-5.12%	[-20.25%, 11.15%]
SVM / Bet365	1/4 Kelly	365	-433.14	-8.57%	[-23.64%, 8.79%]
MLP / Best	Flat 10	333	13.70	0.41%	[-20.08%, 22.07%]
MLP / Best	1/4 Kelly	333	-153.91	-2.59%	[-20.93%, 17.72%]
MLP / Bet365	Flat 10	314	-23.50	-0.75%	[-19.46%, 19.70%]
MLP / Bet365	1/4 Kelly	314	-170.61	-3.09%	[-20.97%, 16.47%]

in cash. Later presentation accounts scale initial capital and fixed stakes by 100; percentage ratios remain unchanged, while monetary amounts must be labelled with their account basis.

3.3 Exploratory team and player comparisons

The rolling extension uses 1,470 common quoted fixtures in four seasons: 368 in 2022/23, 360 in 2023/24, 368 in 2024/25 and 374 in 2025/26. Each fit uses earlier seasons for training, tuning and calibration. These are new redesigns on already examined outcomes; chronology does not make them a fresh confirmatory test.

The earlier source study crosses logistic and market-residual estimators with team, team+xG, team+FPL and combined inputs. Market-residual models start from normalised bookmaker probabilities and learn regularised feature adjustments. The newer study compares logistic and histogram boosting on matched team controls (19 inputs), player form (96), player form plus projected XI/depth (222), and team plus the full player block (235).

Table 11: Selected pooled EPL comparisons on the same 1,470 fixtures. Best learned means lowest observed pooled log loss, not greatest trading capital.

Predictor	Accuracy	Log loss	Brier
Raw market benchmark	54.42%	0.9662	0.5743
Best learned: odds + team/FPL	54.15%	0.9693	0.5751
Team logistic	52.93%	0.9876	0.5889
Player-form logistic	50.07%	1.0218	0.6114
Class frequency	44.29%	1.0700	0.6471

The raw market has the best pooled proper scores among all 19 comparable series. The strongest learned model uses bookmaker probabilities, so it is not an odds-free team/player forecast. All six new player-based variants have worse pooled log loss than their matched same-algorithm team controls. This is a finding about these particular representations, not a general claim that player information has no value.

3.4 Money progression and the complete strategy comparison

The expanded laboratory specifies 116 distinct combinations of forecast, outcome rule, prices and stakes. They are not 116 independent learning algorithms. Every account begins at 100,000. The common 2025/26 chart contains all 116; the uninterrupted four-season chart contains the 88 configurations with comparable

earlier predictions. The 28 unavailable first-study configurations are not backfilled with models that saw those outcomes.

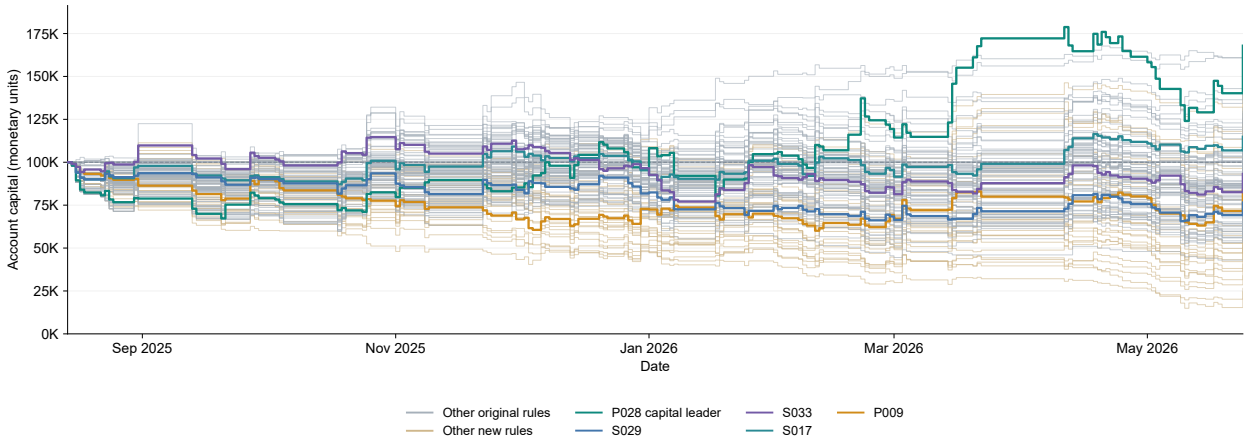


Figure 3: All 116 saved EPL accounts in the common 2025/26 season, each starting at 100,000. A line is one forecast/rule/price/stake combination; overlapping cash paths remain included.

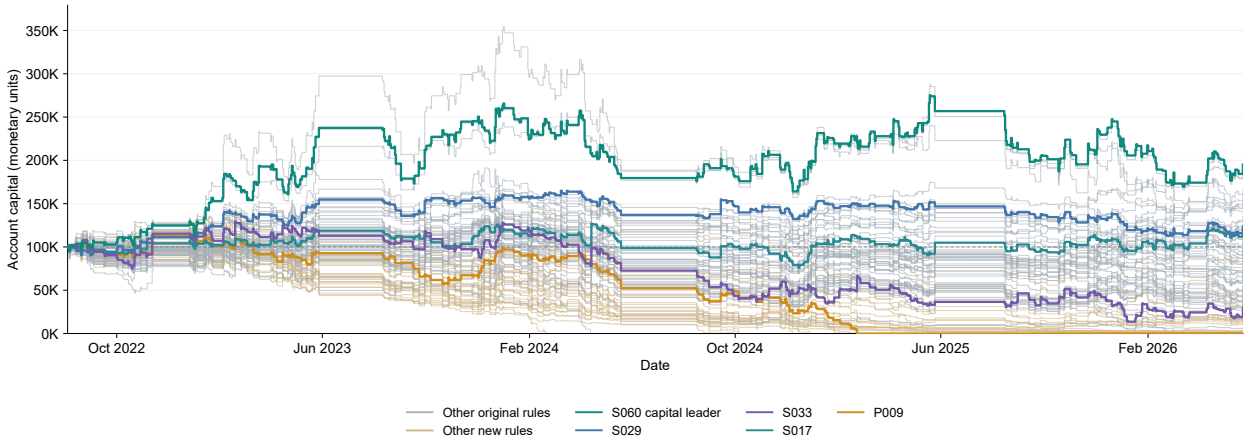


Figure 4: All 88 comparable EPL accounts across four seasons, without annual resets or deposits. Every endpoint reconciles with its saved account metrics.

S029 is the best learned probability model with a flat-stake trading wrapper. S033 is the simplest historical-frequency model, S017 uses team performance, and P009 uses pure player form. Their common 5% value filter and flat 1,000 request make the performance descriptions concrete: 1,000 is 1% of initial capital, not a changing 1% fraction. Requests remain subject to date-start cash allocation.

Among all 116 single-season accounts, 33 profit, 75 lose and eight remain flat. P028 has the highest final 2025/26 capital, 167,925, but loses 83.33% in its uninterrupted four-season account. Among the 88 continuous accounts, 24 profit and 64 lose; 13 exhaust their cash. S060 has the highest ending capital, 195,190, but its conditional ROI interval crosses zero. All 32 new player-study continuous accounts lose money, with 11 reaching zero. Historical ranks are descriptive and do not promote a live betting policy.

Table 12: Representative EPL strategy accounts, uninterrupted 2022/23–2025/26 from 100,000. Expected gain/loss are per-trade model claims.

ID	Bets	Hit	Freq.	Final capital	Stake ROI	G/trade	L/trade	Max DD
S029	1,147	31.04%	78.03%	123,340.00	2.03%	852.95	663.70	31.88%
S033	1,447	23.01%	98.44%	29,760.00	-4.85%	1,739.52	611.96	89.49%
S017	1,241	29.57%	84.42%	119,790.00	1.59%	962.54	648.43	40.35%
P009	944	25.00%	64.22%	0.00	-10.87%	1,098.04	618.22	100.00%
S060	1,186	30.69%	80.68%	195,189.75	2.99%	2,226.45	1,652.54	38.35%
P028	1,328	26.58%	90.34%	16,674.79	-10.67%	628.17	356.67	95.18%

3.5 Prices, operational evidence and joint interpretation

S060's profit is approximately 95,190 at the saved best quotes and 55,952 under a full Bet365 policy replay. Repricing its fixed original exposures instead gives approximately $-30,549$. The latter is a payout attribution; the former recomputes selections and cash sizes. These are different questions, not a pure causal price effect. Quote capture clocks, accepted stake limits and simultaneous fills remain unverified. Mathematical arbitrage flags do not establish executed arbitrage or closing-line value.

The frozen four-season panel flags 4/1,470 mathematical arbitrage candidates (0.2721%), with maximum theoretical margin 0.6272%. Verified executable cases total 0; all 1,470 selected quote sets lack simultaneous clocks. The common 2025/26 panel has 0/374 candidates. These counts answer the arbitrage component without treating flags as earned profit.

The saved operational evidence has a separate October 20 cutoff: four earlier models each classify six of ten resolved fixtures correctly. Forty earlier-model and 80 new-model forecasts concern the same next ten October 23–25 fixtures. This assembly performs no later result settlement, so these figures describe the saved snapshot rather than live October 23 statuses. Missing pre-match quotes and a trade ledger preclude prospective trading ROI.

3.6 Opposite policies after bankruptcy: EPL diagnostic

The 13 zero continuous main accounts comprise S041, S043 and 11 player accounts. Four further zero reference-price replays of S033, S039, S041 and S043 make 17 price contexts across 15 strategy IDs, not 17 independent models. None of the 116 common-season accounts is zero. We also replay the 34 nonbankrupt main fixed-stake accounts as controls; the main comparison therefore covers all 47 fixed-stake accounts, not only selected failures.

In a three-outcome market, "opposite" means covering both outcomes other than the original pick, including the draw. With archived decimal odds O_k, $A = \sum_{k \neq s} 1/O_k$, and total risk request $B = 1,000$, we allocate $B/(O_k A)$ to each complementary outcome. This equalises gross payout at B/A; it does not stake 1,000 on each leg. The synthetic odds $1/A$ can be at or below one, so event occurrence and positive trading profit are not identical. Expected gain/loss sum probability-weighted positive/negative payoffs over all three outcomes, including these loss-making bundles.

We keep the frozen original signal and its entry filter, continue through all 1,470 fixtures after original ruin, and do not impose a new value test on the opposite outcome. Each account starts at 100,000; same-date risks are proportionally scaled to opening cash before settlement, without borrowing or reusing same-date proceeds. One thousand is 1% of initial capital, not a constant fraction of current cash.

At zero added cost, 2 of 13 bankrupt-original complements finish above 100,000, compared with 2 of 34 nonbankrupt controls. These are descriptive, correlated outcomes, not an adjusted hypothesis test. P009's

Table 13: Full-horizon EPL opposite policies from 100,000, main cohort only. "Profit" counts ending balances above 100,000. Costs haircut each fixture's positive net profit by 0%, 2% or 5%; they are scenarios, not observed venue commissions.

Original cohort	Opposite	Cost	N	Profit	Median end	Min end	Max end
Bankrupt originals	Priced complement	0%	13	2	66,059	6,421	116,968
Bankrupt originals	Priced complement	2%	13	2	60,996	2,420	110,249
Bankrupt originals	Priced complement	5%	13	1	53,402	0	100,170
Bankrupt originals	Synthetic lay	0%	13	12	140,580	93,065	165,887
Nonbankrupt controls	Priced complement	0%	34	2	45,269	14,326	107,370
Nonbankrupt controls	Priced complement	2%	34	2	39,888	8,919	106,122
Nonbankrupt controls	Priced complement	5%	34	2	28,549	0	104,251
Nonbankrupt controls	Synthetic lay	0%	34	10	85,532	49,082	148,876

original account ends at zero; its priced complement ends at 92,927, makes 1,384 funded trades from 1,384 frozen signals, and has 29.63% maximum drawdown. Its conditional stake-yield interval is [-3.60%, 2.60%]; these whole-date cashflow intervals omit refitting, resizing and the bankruptcy-based selection.

Across all 17 zero price contexts, the quoted complements profit in 2/17 cases, with median ending capital 62,010 and range 0–116,968. The synthetic same-back-price lay profits in 16/17 cases, with median 140,580. The apparent improvement depends on a different, unobserved price contract; it cannot be attributed to changing only the direction of the bet. No zero-cost priced complement has a conditional 95% stake-yield interval entirely above zero.

A second calculation negates the *recorded original cash flows*; every 100,000-to-zero history then maps to 200,000 by algebra. This truncated-path identity is not the funded complement. A separate synthetic lay fixes liability at 1,000 and uses counterstake $1,000/(O_s - 1)$ at the original back quote. It requires unavailable lay-price assumptions and is not an exchange backtest. Both are labelled diagnostics. The actual complementary bookmaker quotes lack verified simultaneous availability or accepted stakes. Bankruptcy prompted these rules; no inverse is promoted as new confirmed alpha or substituted for the original locked cash policy.

3.7 Final qualitative and quantitative assessment

The shared question is answered by the matched comparisons, not by the largest ex-post account. The original NBA flat capital return favours accuracy selection by 24.05 percentage points (with the SVM reproduction limit); eighth-Kelly favours calibration by 20.11 points while both lose. Across nine annual LR windows calibration wins only three times, with mean returns of -17.34% versus -9.49%. EPL primary policies both hold cash; the secondary flat calibration yield exceeds accuracy by 2.66 points, but its interval includes zero. Better observed ECE therefore does not establish a durable advantage in betting returns. Binary/three-outcome scores, capital returns/stake yields and annual/continuous accounts are not pooled.

The EPL raw market remains the strongest pooled proper-score benchmark, and all six added player variants lose to their matched team controls on log loss. Of 88 continuous exploratory accounts, 24 profit and 64 lose, including 13 bankrupt; the best account's conditional interval includes zero. For the original 84-strategy family only, the saved global bootstrap p-values range from 0.3887–0.4187 versus cash and 0.1457–0.1952 versus the market-favourite baseline across block lengths 1, 3, 5 and 10. These conditional recorded-cashflow tests do not adjust the newer player or inverse families, and no family supplies established profitable alpha.

The strongest contributions are the implemented cross-league comparison, shifted feature construction, ex-

plicit funding, market benchmarks and auditable negative findings. The remaining limits are narrow outcome and season samples, the current-club EPL cohort, approximate/inconsistent source environments, unavailable quote clocks, and post-hoc strategy expansion. Calibration describes probability reliability; betting also depends on prices, the selected subset, costs and exposure. Opposite-policy successes, where observed, test the direction of failed signals and are exploratory. A decisive extension would precommit the model, entry rule, stake budget and evaluation horizon before collecting a fresh cohort of timestamped available quotes. The study supports this discipline rather than a claim of implementable statistical arbitrage.

References

[1] C. Walsh and A. Joshi (2024). Machine learning for sports betting: should model selection be based on accuracy or calibration? *Machine Learning with Applications*, 16, 100539. doi:10.1016/j.mlwa.2024.100539.

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[4] A. Dhillon and P. Housos (2026). *ML Sports Betting: Project Proposal*. Statistical Arbitrage, September 30.

[5] A. Dhillon and P. Housos (2026). *Project data, notebook, frozen models and account ledgers*. Supplied NBA material in `ML Sports`; EPL analysis modules and source-hashed exports in the project repository. Detailed input hashes, evidence classifications and replay reconciliation accompany this report.

A Evidence and capital conventions

The authoritative NBA inputs are the supplied notebook's saved outputs, raw box-score CSV, odds database and stored parameter file. The fixed-parameter audit is labelled separately: logistic results reconcile, SVM results do not. The separate NBA report/presentation is not an additional independent sample. Likewise, the supplied EPL PDF is identical to an earlier generated EPL deck; the reviewed player-model export copies an existing forecast ledger rather than issuing another 80 predictions. Source and output hashes identify every consumed artifact. Current locally edited source documents are preserved.

No joint NBA–EPL account or common profitability average is manufactured. NBA native accounts begin at 10,000; the first EPL experiment begins at 1,000; the expanded EPL account comparison begins at 100,000. Capital return and stake ROI have different denominators. Independent annual NBA restarts are not deposits into a continuous account. Model expected gains and losses are probability-weighted components at the recorded stakes, not average realised payoffs conditional on winning or losing.

B Premier League strategy rules and metrics

The following tables preserve every S001–S084 and P001–P032 definition. Fixed amounts stay fixed as capital changes. Cash fractions and Kelly requests use date-start cash. Same-date requests are scaled together to available cash and any declared date cap, without borrowing or using same-date winnings. "Best" denotes the saved best individual-book quote, not verified executable availability. H/D/A means home/draw/away. Exact machine-readable configurations and complete gain/loss, frequency and risk metrics accompany the report.

B.1 Plain-language capital and entry rules

ID	Model / data	Entry	Stake / cap	Prices
S001	Original SVM forecast	No bets; keep cash	0	No betting
S002	Original SVM forecast	No bets; keep cash	0	No betting
S003	Original SVM forecast	No bets; keep cash	0	No betting
S004	Original SVM forecast	No bets; keep cash	0	No betting
S005	Original SVM forecast	Estimated return > 5%	1,000	Best odds Cash available
S006	Original SVM forecast	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
S007	Original SVM forecast	Estimated return > 5%	1,000	Bet365 Cash available
S008	Original SVM forecast	Estimated return > 5%	1/4 Kelly; ≤2% cash	Bet365 Cash available
S009	Original neural-network forecast	No bets; keep cash	0	No betting
S010	Original neural-network forecast	No bets; keep cash	0	No betting
S011	Original neural-network forecast	No bets; keep cash	0	No betting
S012	Original neural-network forecast	No bets; keep cash	0	No betting

ID	Model / data	Entry	Stake / cap	Prices
S013	Original neural-network forecast	Estimated return > 5%	1,000	Best odds Cash available
S014	Original neural-network forecast	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
S015	Original neural-network forecast	Estimated return > 5%	1,000	Bet365 Cash available
S016	Original neural-network forecast	Estimated return > 5%	1/4 Kelly; ≤2% cash	Bet365 Cash available
S017	Team form	Estimated return > 5%	1,000	Best odds Cash available
S018	Team form	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
S019	Team form + chance quality	Estimated return > 5%	1,000	Best odds Cash available
S020	Team form + chance quality	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
S021	Team form + player form	Estimated return > 5%	1,000	Best odds Cash available
S022	Team form + player form	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
S023	Team + chance quality + player form	Estimated return > 5%	1,000	Best odds Cash available
S024	Team + chance quality + player form	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
S025	Bookmaker odds + team form	Estimated return > 5%	1,000	Best odds Cash available
S026	Bookmaker odds + team form	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
S027	Bookmaker odds + team/chance quality	Estimated return > 5%	1,000	Best odds Cash available
S028	Bookmaker odds + team/chance quality	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
S029	Bookmaker odds + team/player form	Estimated return > 5%	1,000	Best odds Cash available
S030	Bookmaker odds + team/player form	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
S031	Bookmaker odds + team/chance/player form	Estimated return > 5%	1,000	Best odds Cash available

ID	Model / data	Entry	Stake / cap	Prices
S032	Bookmaker odds + team/chance/player form	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
S033	Past home/draw/away frequencies	Estimated return > 5%	1,000	Best odds Cash available
S034	Past home/draw/away frequencies	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
S035	Bookmaker probabilities	Estimated return > 5%	1,000	Best odds Cash available
S036	Bookmaker probabilities	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
S037	Calibrated bookmaker probabilities	Estimated return > 5%	1,000	Best odds Cash available
S038	Calibrated bookmaker probabilities	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
S039	Bookmaker probabilities	Always home win No return filter	1,000	Best odds Cash available
S040	Bookmaker probabilities	Always draw No return filter	1,000	Best odds Cash available
S041	Bookmaker probabilities	Always away win No return filter	1,000	Best odds Cash available
S042	Bookmaker probabilities	Bookmaker's most likely result No return filter	1,000	Best odds Cash available
S043	Bookmaker probabilities	Bookmaker's least likely result No return filter	1,000	Best odds Cash available
S044	Team + chance quality + player form	Model's most likely result No return filter	1,000	Best odds Cash available
S045	Bookmaker odds + team/chance/player form	Model's most likely result No return filter	1,000	Best odds Cash available
S046	Average of four team/player forecasts	Estimated return > 5%	1,000	Best odds Cash available
S047	Average of four team/player forecasts	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
S048	Average of eight learned forecasts	Estimated return > 5%	1,000	Best odds Cash available
S049	Average of eight learned forecasts	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
S050	Half combined forecast, half bookmaker	Estimated return > 5%	1,000	Best odds Cash available

ID	Model / data	Entry	Stake / cap	Prices
S051	Half combined forecast, half bookmaker	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
S052	Team + chance quality + player form	Estimated return > 5%	1% of current cash	Best odds Cash available
S053	Bookmaker odds + team/chance/player form	Estimated return > 5%	1% of current cash	Best odds Cash available
S054	Average of eight learned forecasts	Estimated return > 5%	1% of current cash	Best odds Cash available
S055	Team + chance quality + player form	Estimated return > 5%	1/8 Kelly; ≤1% cash	Best odds Cash available
S056	Team + chance quality + player form	Estimated return > 5%	1/2 Kelly; ≤2% cash	Best odds Cash available
S057	Bookmaker odds + team/chance/player form	Estimated return > 5%	1/8 Kelly; ≤1% cash	Best odds Cash available
S058	Bookmaker odds + team/chance/player form	Estimated return > 5%	1/2 Kelly; ≤2% cash	Best odds Cash available
S059	Team + chance quality + player form	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Date cap 10% cash
S060	Bookmaker odds + team/chance/player form	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Date cap 10% cash
S061	Average of eight learned forecasts	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Date cap 10% cash
S062	Average of eight learned forecasts	Estimated return > 5% Positive return: 6/8 models	1,000	Best odds Cash available
S063	Average of eight learned forecasts	Estimated return > 5% Positive return: 6/8 models	1/4 Kelly; ≤2% cash	Best odds Cash available
S064	Average of eight learned forecasts	Estimated return > 5% Only H	1,000	Best odds Cash available
S065	Average of eight learned forecasts	Estimated return > 5% Only D	1,000	Best odds Cash available
S066	Average of eight learned forecasts	Estimated return > 5% Only A	1,000	Best odds Cash available
S067	Team + chance quality + player form	Estimated return > 10%	1,000	Best odds Cash available
S068	Bookmaker odds + team/chance/player form	Estimated return > 10%	1,000	Best odds Cash available
S069	Average of eight learned forecasts	Estimated return > 10%	1,000	Best odds Cash available

ID	Model / data	Entry	Stake / cap	Prices
S070	Team + chance quality + player form	Estimated return > 5% Odds 1.5–5.0	1,000	Best odds Cash available
S071	Bookmaker odds + team/chance/player form	Estimated return > 5% Odds 1.5–5.0	1,000	Best odds Cash available
S072	Average of eight learned forecasts	Estimated return > 5% Odds 1.5–5.0	1,000	Best odds Cash available
S073	Original SVM forecast	Estimated return > 0%	1,000	Best odds Cash available
S074	Original SVM forecast	Estimated return > 2.5%	1,000	Best odds Cash available
S075	Original SVM forecast	Estimated return > 7.5%	1,000	Best odds Cash available
S076	Original SVM forecast	Estimated return > 10%	1,000	Best odds Cash available
S077	Original SVM forecast	Estimated return > 15%	1,000	Best odds Cash available
S078	Original SVM forecast	Estimated return > 20%	1,000	Best odds Cash available
S079	Original neural-network forecast	Estimated return > 0%	1,000	Best odds Cash available
S080	Original neural-network forecast	Estimated return > 2.5%	1,000	Best odds Cash available
S081	Original neural-network forecast	Estimated return > 7.5%	1,000	Best odds Cash available
S082	Original neural-network forecast	Estimated return > 10%	1,000	Best odds Cash available
S083	Original neural-network forecast	Estimated return > 15%	1,000	Best odds Cash available
S084	Original neural-network forecast	Estimated return > 20%	1,000	Best odds Cash available
P001	Team form (logistic)	Estimated return > 5%	1,000	Best odds Cash available
P002	Team form (logistic)	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
P003	Team form (logistic)	Estimated return > 5%	1,000	Bet365 Cash available
P004	Team form (logistic)	Estimated return > 5%	1/4 Kelly; ≤2% cash	Bet365 Cash available

ID	Model / data	Entry	Stake / cap	Prices
P005	Team form (boosted trees)	Estimated return > 5%	1,000	Best odds Cash available
P006	Team form (boosted trees)	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
P007	Team form (boosted trees)	Estimated return > 5%	1,000	Bet365 Cash available
P008	Team form (boosted trees)	Estimated return > 5%	1/4 Kelly; ≤2% cash	Bet365 Cash available
P009	Player form (logistic)	Estimated return > 5%	1,000	Best odds Cash available
P010	Player form (logistic)	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
P011	Player form (logistic)	Estimated return > 5%	1,000	Bet365 Cash available
P012	Player form (logistic)	Estimated return > 5%	1/4 Kelly; ≤2% cash	Bet365 Cash available
P013	Player form (boosted trees)	Estimated return > 5%	1,000	Best odds Cash available
P014	Player form (boosted trees)	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
P015	Player form (boosted trees)	Estimated return > 5%	1,000	Bet365 Cash available
P016	Player form (boosted trees)	Estimated return > 5%	1/4 Kelly; ≤2% cash	Bet365 Cash available
P017	Player form + estimated XI/depth (logistic)	Estimated return > 5%	1,000	Best odds Cash available
P018	Player form + estimated XI/depth (logistic)	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
P019	Player form + estimated XI/depth (logistic)	Estimated return > 5%	1,000	Bet365 Cash available
P020	Player form + estimated XI/depth (logistic)	Estimated return > 5%	1/4 Kelly; ≤2% cash	Bet365 Cash available
P021	Player form + estimated XI/depth (trees)	Estimated return > 5%	1,000	Best odds Cash available
P022	Player form + estimated XI/depth (trees)	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
P023	Player form + estimated XI/depth (trees)	Estimated return > 5%	1,000	Bet365 Cash available

ID	Model / data	Entry	Stake / cap	Prices
P024	Player form + estimated XI/depth (trees)	Estimated return > 5%	1/4 Kelly; ≤2% cash	Bet365 Cash available
P025	Team + player form/XI/depth (logistic)	Estimated return > 5%	1,000	Best odds Cash available
P026	Team + player form/XI/depth (logistic)	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
P027	Team + player form/XI/depth (logistic)	Estimated return > 5%	1,000	Bet365 Cash available
P028	Team + player form/XI/depth (logistic)	Estimated return > 5%	1/4 Kelly; ≤2% cash	Bet365 Cash available
P029	Team + player form/XI/depth (trees)	Estimated return > 5%	1,000	Best odds Cash available
P030	Team + player form/XI/depth (trees)	Estimated return > 5%	1/4 Kelly; ≤2% cash	Best odds Cash available
P031	Team + player form/XI/depth (trees)	Estimated return > 5%	1,000	Bet365 Cash available
P032	Team + player form/XI/depth (trees)	Estimated return > 5%	1/4 Kelly; ≤2% cash	Bet365 Cash available

B.2 Complete common-season strategy statistics

Every row below starts at 100,000 and uses the same 374 priced 2025/26 fixtures. Expected gain/loss are per placed trade. No-trade conditional quantities are shown as unavailable, rather than fabricated as zero.

ID	Bets	Hit	Freq.	G/trade	L/trade	Final capital	ROI	Max DD
S001	0	–	0.00%	–	–	100,000.00	–	0.00%
S002	0	–	0.00%	–	–	100,000.00	–	0.00%
S003	0	–	0.00%	–	–	100,000.00	–	0.00%
S004	0	–	0.00%	–	–	100,000.00	–	0.00%
S005	371	26.68%	99.20%	1,173.68	631.47	91,670.00	-2.25%	27.74%
S006	371	26.68%	99.20%	1,748.01	905.17	62,295.04	-7.01%	49.95%
S007	365	26.85%	97.59%	1,087.19	625.18	81,330.00	-5.12%	28.60%
S008	365	26.85%	97.59%	1,545.18	854.83	56,685.92	-8.57%	51.21%
S009	0	–	0.00%	–	–	100,000.00	–	0.00%
S010	0	–	0.00%	–	–	100,000.00	–	0.00%
S011	0	–	0.00%	–	–	100,000.00	–	0.00%
S012	0	–	0.00%	–	–	100,000.00	–	0.00%
S013	333	27.03%	89.04%	1,240.24	631.66	101,370.00	0.41%	28.11%
S014	333	27.03%	89.04%	2,295.30	1,107.73	84,608.55	-2.59%	54.44%
S015	314	27.71%	83.96%	1,160.52	628.63	97,650.00	-0.75%	29.38%
S016	314	27.71%	83.96%	2,101.94	1,084.41	82,939.09	-3.09%	51.98%
S017	306	27.78%	81.82%	1,034.32	690.01	114,870.00	4.86%	18.00%
S018	306	27.78%	81.82%	1,430.25	903.46	97,279.01	-0.67%	35.79%
S019	304	29.28%	81.28%	1,019.10	689.70	142,270.00	13.90%	15.10%
S020	304	29.28%	81.28%	1,603.27	1,036.33	118,763.06	4.04%	31.36%

ID	Bets	Hit	Freq.	G/trade	L/trade	Final capital	ROI	Max DD
S021	314	26.11%	83.96%	1,035.00	692.84	105,380.00	1.71%	21.25%
S022	314	26.11%	83.96%	1,231.39	782.29	64,151.32	-9.94%	48.86%
S023	321	27.10%	85.83%	1,010.98	689.58	112,730.00	3.97%	22.14%
S024	321	27.10%	85.83%	1,293.70	838.41	70,888.52	-7.33%	47.23%
S025	250	24.40%	66.84%	893.26	720.60	75,660.00	-9.74%	34.22%
S026	250	24.40%	66.84%	867.71	669.15	83,812.42	-6.63%	25.64%
S027	259	27.03%	69.25%	896.62	717.47	110,760.00	4.15%	19.50%
S028	259	27.03%	69.25%	957.16	735.52	100,888.45	0.32%	19.13%
S029	277	25.27%	74.06%	897.20	709.19	76,550.00	-8.47%	34.33%
S030	277	25.27%	74.06%	915.50	696.55	74,738.60	-8.89%	32.58%
S031	289	24.22%	77.27%	900.22	710.69	69,770.00	-10.46%	44.57%
S032	289	24.22%	77.27%	898.72	680.11	75,980.20	-8.32%	33.95%
S033	368	25.00%	98.40%	1,511.12	613.81	93,220.00	-1.84%	32.78%
S034	368	25.00%	98.40%	2,505.09	987.80	67,519.28	-5.46%	57.66%
S035	62	6.45%	16.58%	982.31	861.00	62,500.00	-60.48%	40.00%
S036	62	6.45%	16.58%	357.20	308.84	88,934.54	-49.40%	13.93%
S037	91	6.59%	24.33%	970.99	844.61	43,000.00	-62.64%	57.00%
S038	91	6.59%	24.33%	385.79	329.87	80,836.96	-53.73%	21.26%
S039	374	42.51%	100.00%	537.20	562.74	72,180.00	-7.44%	44.37%
S040	374	27.54%	100.00%	729.43	756.07	141,880.00	11.20%	32.74%
S041	374	29.95%	100.00%	676.98	681.19	75,360.00	-6.59%	40.75%
S042	374	48.66%	100.00%	443.06	478.23	61,840.00	-10.20%	41.23%
S043	374	22.19%	100.00%	784.06	780.78	83,310.00	-4.46%	38.25%
S044	374	48.93%	100.00%	468.52	514.06	77,300.00	-6.07%	31.07%
S045	374	48.40%	100.00%	451.15	489.01	62,820.00	-9.94%	41.77%
S046	304	28.62%	81.28%	1,027.68	693.72	128,390.00	9.34%	17.78%
S047	304	28.62%	81.28%	1,569.05	1,010.16	101,520.08	0.34%	36.06%
S048	289	28.37%	77.27%	963.99	716.17	131,910.00	11.04%	20.55%
S049	289	28.37%	77.27%	1,147.59	812.53	96,316.90	-1.10%	34.23%
S050	239	22.59%	63.90%	956.46	746.18	91,520.00	-3.55%	28.95%
S051	239	22.59%	63.90%	793.30	589.60	59,111.21	-21.19%	43.87%
S052	321	27.10%	85.83%	990.69	676.42	105,964.88	1.90%	23.80%
S053	289	24.22%	77.27%	688.46	542.44	70,723.65	-13.25%	37.65%
S054	289	28.37%	77.27%	1,028.92	765.50	129,697.99	9.63%	19.30%
S055	321	27.10%	85.83%	738.16	479.10	87,620.08	-5.46%	26.12%
S056	321	27.10%	85.83%	1,718.96	1,155.98	93,017.68	-1.29%	41.49%
S057	289	24.22%	77.27%	503.50	381.48	88,673.78	-6.99%	17.61%
S058	289	24.22%	77.27%	1,079.15	834.37	63,021.45	-10.66%	49.37%
S059	321	27.10%	85.83%	1,287.68	834.50	69,280.57	-7.78%	47.04%
S060	289	24.22%	77.27%	898.72	680.11	75,980.20	-8.32%	33.95%
S061	289	28.37%	77.27%	1,145.65	811.06	95,751.43	-1.27%	34.23%
S062	270	28.15%	72.19%	975.01	716.27	125,710.00	9.52%	20.05%
S063	270	28.15%	72.19%	1,172.38	823.53	93,983.80	-1.89%	33.60%
S064	116	31.90%	31.02%	929.65	674.71	107,900.00	6.81%	13.37%
S065	135	32.59%	36.10%	896.42	740.92	158,010.00	42.97%	9.83%
S066	130	19.23%	34.76%	1,032.62	759.43	85,660.00	-11.03%	23.49%
S067	264	25.38%	70.59%	1,075.87	700.54	111,450.00	4.34%	20.45%
S068	200	21.50%	53.48%	962.06	721.36	71,500.00	-14.25%	37.18%
S069	222	24.77%	59.36%	1,028.87	728.19	112,670.00	5.71%	21.00%
S070	256	33.20%	68.45%	857.56	655.54	142,810.00	16.72%	13.00%
S071	222	31.08%	59.36%	816.10	667.18	115,150.00	6.82%	16.30%
S072	217	35.94%	58.02%	833.94	674.32	161,240.00	28.22%	12.00%
S073	374	26.74%	100.00%	1,170.29	632.09	92,220.00	-2.08%	28.00%
S074	374	26.74%	100.00%	1,170.29	632.09	92,220.00	-2.08%	28.00%
S075	365	26.58%	97.59%	1,181.53	631.43	91,870.00	-2.23%	28.54%

ID	Bets	Hit	Freq.	G/trade	L/trade	Final capital	ROI	Max DD
S076	349	26.65%	93.32%	1,201.85	630.70	93,170.00	-1.96%	27.81%
S077	312	25.32%	83.42%	1,258.99	635.05	88,950.00	-3.54%	27.72%
S078	276	26.09%	73.80%	1,317.97	634.90	104,400.00	1.59%	18.87%
S079	364	26.37%	97.33%	1,194.85	635.56	89,910.00	-2.77%	36.93%
S080	354	26.84%	94.65%	1,208.70	633.98	96,810.00	-0.90%	31.65%
S081	318	27.36%	85.03%	1,263.27	628.83	103,370.00	1.06%	24.76%
S082	302	27.15%	80.75%	1,294.95	631.49	107,220.00	2.39%	23.33%
S083	275	25.82%	73.53%	1,346.59	630.29	96,530.00	-1.26%	32.13%
S084	255	23.92%	68.18%	1,388.55	629.78	87,650.00	-4.84%	35.93%
P001	310	24.52%	82.89%	1,059.58	702.17	91,620.00	-2.70%	32.39%
P002	310	24.52%	82.89%	1,174.18	730.93	69,816.37	-9.21%	47.39%
P003	274	24.82%	73.26%	1,006.81	699.86	103,900.00	1.42%	32.87%
P004	274	24.82%	73.26%	1,175.72	767.06	90,961.46	-2.96%	42.89%
P005	318	24.21%	85.03%	1,050.63	690.45	80,610.00	-6.10%	39.47%
P006	318	24.21%	85.03%	1,146.63	703.87	49,066.59	-15.47%	59.73%
P007	288	25.69%	77.01%	997.49	686.05	93,180.00	-2.37%	21.97%
P008	288	25.69%	77.01%	1,156.47	748.10	59,110.00	-12.79%	50.67%
P009	348	24.14%	93.05%	1,122.19	678.28	83,820.00	-4.65%	39.90%
P010	348	24.14%	93.05%	1,167.70	663.00	52,544.05	-13.78%	63.88%
P011	332	24.70%	88.77%	1,060.23	672.08	89,810.00	-3.07%	47.70%
P012	332	24.70%	88.77%	1,101.19	655.41	64,919.02	-10.70%	59.54%
P013	323	22.29%	86.36%	1,153.99	678.16	68,460.00	-9.76%	52.37%
P014	323	22.29%	86.36%	1,333.67	740.56	40,571.80	-16.63%	67.04%
P015	305	22.95%	81.55%	1,077.69	675.10	73,840.00	-8.58%	42.27%
P016	305	22.95%	81.55%	1,280.96	758.63	51,816.24	-13.86%	56.67%
P017	359	25.63%	95.99%	1,169.25	667.00	107,320.00	2.04%	19.32%
P018	359	25.63%	95.99%	1,780.72	980.45	92,632.56	-1.39%	35.98%
P019	349	27.79%	93.32%	1,099.10	661.60	131,580.00	9.05%	16.10%
P020	349	27.79%	93.32%	2,011.10	1,168.39	141,923.84	6.77%	28.29%
P021	334	19.46%	89.30%	1,147.31	679.67	26,520.00	-22.00%	85.81%
P022	334	19.46%	89.30%	1,117.92	612.44	24,500.49	-24.87%	82.21%
P023	309	22.01%	82.62%	1,082.41	678.78	60,440.00	-12.80%	58.51%
P024	309	22.01%	82.62%	1,126.02	659.02	38,006.18	-20.46%	73.72%
P025	358	26.82%	95.72%	1,134.64	665.45	120,950.00	5.85%	20.85%
P026	358	26.82%	95.72%	1,833.47	1,035.36	122,960.82	4.09%	36.97%
P027	345	28.70%	92.25%	1,072.34	662.61	144,750.00	12.97%	15.35%
P028	345	28.70%	92.25%	2,057.39	1,218.96	167,925.38	10.62%	32.79%
P029	340	22.65%	90.91%	1,064.49	676.14	50,290.00	-14.62%	67.52%
P030	340	22.65%	90.91%	1,012.85	601.23	33,915.29	-21.49%	74.08%
P031	308	25.32%	82.35%	1,012.93	673.44	82,050.00	-5.83%	36.06%
P032	308	25.32%	82.35%	1,129.58	708.34	55,958.98	-13.35%	57.43%

C Bankrupt-policy opposite audit

The 17 EPL contexts below include the 13 main zero accounts and four additional reference-price policy replays. Repeated IDs with different prices are different contexts. The two NBA zero annual accounts are shown separately: the audit therefore covers 19 zero-account cases across the leagues, without treating them as independent tests. Exact recorded-profit negation always gives 200,000 for EPL (20,000 for NBA) by arithmetic; this is not an attainable-return estimate or a funded trade ledger. On the recorded truncated paths, 17/17 EPL mirrors pass the hypothetical same-date liability-reserve check; the maximum additional opening-cash requirement is 0. Thus extra funding is not the observed obstruction for these exact recorded mirrors. The absent opposing contract and its unobserved price remain the obstructions to claiming an executable mirror; funding feasibility alone does not establish availability.

Table 16: All 17 EPL zero-original price contexts; every funded inverse starts at 100,000. Original end capital is zero. The first four numeric columns are ending capital; cost haircuts positive fixture net profit. Mirror gap is additional opening cash needed to reserve the hypothetical exact-counterparty liability on at least one date; it is not the funded complement's cash requirement.

ID	Quotes	Context	Comp. 0%	Comp. 2%	Comp. 5%	Lay 0%	Mirror gap
S041	best	Main	83,274	74,388	61,060	140,580	0
S043	best	Main	63,774	58,478	50,535	107,026	0
P009	best	Main	92,927	86,268	76,280	138,377	0
P011	reference	Main	49,178	43,833	35,816	145,073	0
P013	best	Main	94,727	88,593	79,391	138,138	0
P015	reference	Main	66,059	60,996	53,402	159,942	0
P019	reference	Main	20,121	14,677	6,510	118,747	0
P021	best	Main	115,263	108,878	99,300	160,426	0
P023	reference	Main	70,366	65,216	57,491	165,887	0
P025	best	Main	48,758	42,443	32,970	93,065	0
P027	reference	Main	6,421	2,420	0	101,274	0
P029	best	Main	116,968	110,249	100,170	162,936	0
P031	reference	Main	62,010	56,767	48,903	155,770	0
S033	reference	Ref. replay	24,426	19,634	12,446	120,882	0
S039	reference	Ref. replay	0	0	0	176,440	0
S041	reference	Ref. replay	39,139	31,133	19,124	168,948	0
S043	reference	Ref. replay	26,653	22,097	15,262	124,838	0

C.1 Risk and payoff rules

All EPL inverses use the original frozen qualification over the full 1,470-fixture horizon. Complement requests split a total 1,000 between the two alternative outcomes in reciprocal-odds proportions. Synthetic lay requests reserve total liability 1,000 and compute the counterstake from the back quote; they do not assume observed exchange liquidity. Risks are scaled together to date-start cash. Cost sensitivity applies to a fixture's positive *net bundle profit*, with no rebate on losses, and cash sizes are replayed for each scenario. Thus the additional haircut does not double count a winning leg while ignoring its losing complementary leg. Bookmaker vig is already embedded in the archived odds.

Hit rate reports occurrence of the complementary event. A bundle can still lose if its synthetic odds do not exceed one; the CSV also preserves the fraction of bundles earning positive net profit. Expected components use the original forecast probabilities and signed outcome payoffs, not the empirical frequency of wins. The

original model need not endorse its contrarian trade. Full CSVs additionally contain turnover, realised payoff components, funding checks and source/context identifiers.

Table 17: Complete funded inverse statistics for the 17 zero contexts. Expected payoff components are model-implied monetary units per placed trade; frequency is placed trades divided by the 1,470 opportunities. DD is maximum date-close drawdown.

ID	Quotes	Mode	Cost	Signals	Bets	Hit	Freq.	Exp. gain	Exp. loss	Risk yield	DD
S041	best	Complement	0%	1,470	1,470	68.64%	100.00%	301.2	323.0	-1.14%	39.24%
S041	best	Complement	2%	1,470	1,470	68.64%	100.00%	295.2	323.0	-1.74%	45.76%
S041	best	Complement	5%	1,470	1,470	68.64%	100.00%	286.1	323.0	-2.65%	55.75%
S041	best	Lay	0%	1,470	1,470	68.64%	100.00%	339.8	323.0	2.76%	15.40%
S043	best	Complement	0%	1,470	1,470	79.52%	100.00%	180.8	207.3	-2.46%	36.38%
S043	best	Complement	2%	1,470	1,470	79.52%	100.00%	177.2	207.3	-2.82%	41.66%
S043	best	Complement	5%	1,470	1,470	79.52%	100.00%	171.8	207.3	-3.36%	49.58%
S043	best	Lay	0%	1,470	1,470	79.52%	100.00%	210.3	207.3	0.48%	10.61%
P009	best	Complement	0%	1,384	1,384	75.43%	94.15%	199.2	354.8	-0.51%	29.63%
P009	best	Complement	2%	1,384	1,384	75.43%	94.15%	195.2	354.8	-0.99%	31.35%
P009	best	Complement	5%	1,384	1,384	75.43%	94.15%	189.3	354.8	-1.71%	34.01%
P009	best	Lay	0%	1,384	1,384	75.43%	94.15%	227.0	354.8	2.77%	19.72%
P011	reference	Complement	0%	1,290	1,290	75.35%	87.76%	170.5	356.1	-3.94%	50.93%
P011	reference	Complement	2%	1,290	1,290	75.35%	87.76%	167.1	356.1	-4.35%	56.26%
P011	reference	Complement	5%	1,290	1,290	75.35%	87.76%	162.0	356.1	-4.98%	64.26%
P011	reference	Lay	0%	1,290	1,290	75.35%	87.76%	232.5	356.1	3.49%	13.87%
P013	best	Complement	0%	1,351	1,351	76.91%	91.90%	188.8	342.0	-0.39%	21.94%
P013	best	Complement	2%	1,351	1,351	76.91%	91.90%	185.0	342.0	-0.84%	25.00%
P013	best	Complement	5%	1,351	1,351	76.91%	91.90%	179.3	342.0	-1.53%	30.13%
P013	best	Lay	0%	1,351	1,351	76.91%	91.90%	216.0	342.0	2.82%	13.81%
P015	reference	Complement	0%	1,276	1,276	77.51%	86.80%	163.1	344.6	-2.66%	34.18%
P015	reference	Complement	2%	1,276	1,276	77.51%	86.80%	159.8	344.6	-3.06%	39.14%
P015	reference	Complement	5%	1,276	1,276	77.51%	86.80%	154.9	344.6	-3.65%	46.71%
P015	reference	Lay	0%	1,276	1,276	77.51%	86.80%	224.4	344.6	4.70%	9.65%
P019	reference	Complement	0%	1,360	1,360	74.12%	92.52%	167.6	365.5	-5.87%	80.38%
P019	reference	Complement	2%	1,360	1,360	74.12%	92.52%	164.3	365.5	-6.27%	85.67%
P019	reference	Complement	5%	1,360	1,360	74.12%	92.52%	159.3	365.5	-6.87%	93.63%
P019	reference	Lay	0%	1,360	1,360	74.12%	92.52%	228.8	365.4	1.38%	15.47%
P021	best	Complement	0%	1,370	1,370	77.81%	93.20%	191.2	344.9	1.11%	16.20%
P021	best	Complement	2%	1,370	1,370	77.81%	93.20%	187.4	344.9	0.65%	17.94%
P021	best	Complement	5%	1,370	1,370	77.81%	93.20%	181.6	344.9	-0.05%	21.81%
P021	best	Lay	0%	1,370	1,370	77.81%	93.20%	218.5	344.9	4.41%	7.57%
P023	reference	Complement	0%	1,287	1,287	77.70%	87.55%	164.1	346.5	-2.30%	29.82%
P023	reference	Complement	2%	1,287	1,287	77.70%	87.55%	160.8	346.5	-2.70%	34.93%
P023	reference	Complement	5%	1,287	1,287	77.70%	87.55%	155.9	346.5	-3.30%	42.63%
P023	reference	Lay	0%	1,287	1,287	77.70%	87.55%	225.6	346.4	5.12%	5.84%
P025	best	Complement	0%	1,402	1,402	73.82%	95.37%	194.8	359.4	-3.65%	52.02%
P025	best	Complement	2%	1,402	1,402	73.82%	95.37%	190.9	359.4	-4.11%	58.18%
P025	best	Complement	5%	1,402	1,402	73.82%	95.37%	185.1	359.4	-4.78%	67.45%
P025	best	Lay	0%	1,402	1,402	73.82%	95.37%	222.5	359.4	-0.49%	19.98%
P027	reference	Complement	0%	1,328	1,328	73.42%	90.34%	169.3	362.6	-7.05%	93.61%
P027	reference	Complement	2%	1,328	1,328	73.42%	90.34%	165.5	361.8	-7.37%	97.59%
P027	reference	Complement	5%	1,328	1,283	73.34%	87.28%	159.1	359.0	-7.90%	100.00%
P027	reference	Lay	0%	1,328	1,328	73.42%	90.34%	231.1	362.5	0.10%	20.92%
P029	best	Complement	0%	1,346	1,346	76.30%	91.56%	202.2	348.5	1.26%	13.98%
P029	best	Complement	2%	1,346	1,346	76.30%	91.56%	198.1	348.5	0.76%	16.12%
P029	best	Complement	5%	1,346	1,346	76.30%	91.56%	192.1	348.5	0.01%	19.36%
P029	best	Lay	0%	1,346	1,346	76.30%	91.56%	230.5	348.5	4.68%	10.48%
P031	reference	Complement	0%	1,234	1,234	75.69%	83.95%	172.5	348.9	-3.08%	38.47%
P031	reference	Complement	2%	1,234	1,234	75.69%	83.95%	169.0	348.9	-3.50%	43.62%
P031	reference	Complement	5%	1,234	1,234	75.69%	83.95%	163.9	348.9	-4.14%	51.36%
P031	reference	Lay	0%	1,234	1,234	75.69%	83.95%	235.5	348.8	4.52%	11.25%
S033	reference	Complement	0%	1,409	1,409	77.64%	95.85%	142.3	387.4	-5.36%	75.63%

ID	Quotes	Mode	Cost	Signals	Bets	Hit	Freq.	Exp. gain	Exp. loss	Risk yield	DD
S033	reference	Complement	2%	1,409	1,409	77.64%	95.85%	139.5	387.4	-5.70%	80.41%
S033	reference	Complement	5%	1,409	1,409	77.64%	95.85%	135.2	387.4	-6.21%	87.58%
S033	reference	Lay	0%	1,409	1,409	77.64%	95.85%	197.3	387.4	1.48%	14.85%
S039	reference	Complement	0%	1,470	1,194	54.52%	81.22%	368.3	416.7	-8.91%	100.00%
S039	reference	Complement	2%	1,470	827	53.08%	56.26%	369.8	426.6	-12.54%	100.00%
S039	reference	Complement	5%	1,470	788	53.68%	53.61%	338.6	403.0	-13.91%	100.00%
S039	reference	Lay	0%	1,470	1,470	55.71%	100.00%	513.3	442.1	5.20%	22.19%
S041	reference	Complement	0%	1,470	1,470	68.64%	100.00%	271.5	323.1	-4.14%	73.18%
S041	reference	Complement	2%	1,470	1,470	68.64%	100.00%	266.0	323.1	-4.68%	79.55%
S041	reference	Complement	5%	1,470	1,470	68.64%	100.00%	257.9	323.1	-5.50%	89.26%
S041	reference	Lay	0%	1,470	1,470	68.64%	100.00%	359.7	323.0	4.69%	11.89%
S043	reference	Complement	0%	1,470	1,470	79.52%	100.00%	155.7	207.4	-4.99%	73.41%
S043	reference	Complement	2%	1,470	1,470	79.52%	100.00%	152.6	207.4	-5.30%	77.95%
S043	reference	Complement	5%	1,470	1,470	79.52%	100.00%	147.9	207.4	-5.76%	84.77%
S043	reference	Lay	0%	1,470	1,470	79.52%	100.00%	222.2	207.3	1.69%	8.82%

Evidence: `data/epl_inverse_metrics.csv;data/epl_identity_diagnostics.csv;data/epl_cohort_summary.json`

C.2 NBA zero-account details

All rows use the 2017/18 rolling logistic-only reconstruction with a fresh 10,000-unit account. The original requests 100; prefix reversals request the recorded original amounts, including a final cash-clamped stake. Full-horizon reversals request 100 per frozen original signal. "Expected" means the original model's probability-weighted payoff for the contrarian outcome.

Branch	Mode	Bets	End	Hit	Exp. gain	Exp. loss	Frequency	Drawdown
Acc.	Opposite full	884	8,785.30	63.35%	27.99	47.91	82.23%	50.76%
Acc.	Opposite prefix	868	8,155.98	62.90%	28.09	47.81	80.74%	50.76%
Acc.	Original	868	0.00	37.10%	78.66	52.13	80.74%	100.00%
Cal.	Opposite full	915	11,148.07	65.68%	27.06	46.67	85.12%	29.17%
Cal.	Opposite prefix	704	9,420.33	63.07%	28.42	47.47	65.49%	29.17%
Cal.	Original	704	0.00	36.93%	75.79	52.45	65.49%	100.00%

Evidence: `data/nba_inverse_metrics.csv;data/nba_conditional_uncertainty.csv`